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Designing Applications for Gossip Routed Decentralized Networks

Programų kūrimas „Gossip“ protokolu grindžiamiems betėvams tinklams

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Abstract

In decentralized networks, traditional epidemic routing algorithms like the Gossip protocol offer high fault-tolerance but suffer from extreme message redundancy. This redundancy consumes significant bandwidth and unnecessarily exposes large cryptographic payloads to untrusted intermediate nodes, forcing a compromise between network overhead and data confidentiality. To resolve this limitation, this study proposes a novel architecture that completely decouples the cryptographic payload from the gossip routing layer by utilizing the InterPlanetary File System (IPFS) for secure, off-path storage. Consequently, the data transmitted through the network is limited to a lightweight, fixed-size Content Identifier (CID). The system optimizes efficiency and security by applying dynamic Shamir’s Secret Sharing (SSS) exclusively to the CID, distributing the fragments via an Advanced Adaptive Gossiping (AAG) protocol, and verifying fragment integrity hop-by-hop using a Hash-Based Verifiable Secret Sharing (VSS) mechanism.

To evaluate the resilience of this architecture, distributed Docker-based simulations were conducted across linear, ring, mesh, and bottleneck topologies ranging from 10 to 20 nodes. A primary focus of the evaluation was a probabilistic revocation mechanism, which utilizes an artificially accelerated anti-rumor command designed to overtake and neutralize in-transit fragments. Experimental results demonstrated that this fast-lane revocation successfully prevents target nodes from reconstructing the secret, provided the network possesses sufficient propagation depth. However, the evaluation also exposed a critical flaw in dense configurations, identified as the Multipath Convergence Vulnerability. In highly connected or bottlenecked network structures, deterministic routing filters organically funnel distinct cryptographic lanes into single intermediate nodes, allowing them to unintentionally accumulate the required threshold and reconstruct the secret prematurely. These findings underline the necessity for future systems to integrate dynamic bottleneck detection, decentralized reputation matrices, and metadata obfuscation to fully secure decentralized environments.

Resumo

Em redes descentralizadas, algoritmos tradicionais de roteamento epidêmico, como o protocolo Gossip, oferecem alta tolerância a falhas, mas sofrem com a extrema redundância de mensagens. Essa redundância consome uma largura de banda significativa e expõe desnecessariamente grandes cargas criptográficas (payloads) a nós intermediários não confiáveis, forçando um compromisso entre a sobrecarga da rede e a confidencialidade dos dados. Para resolver essa limitação, este estudo propõe uma arquitetura inovadora que desvincula completamente a carga criptográfica da camada de roteamento gossip, utilizando o InterPlanetary File System (IPFS) para armazenamento seguro fora do caminho de roteamento (off-path). Consequentemente, os dados transmitidos através da rede limitam-se a um Identificador de Conteúdo (CID) leve e de tamanho fixo. O sistema otimiza a eficiência e a segurança ao aplicar o Compartilhamento de Segredos de Shamir (SSS) dinâmico exclusivamente ao CID, distribuindo os fragmentos através de um protocolo de Advanced Adaptive Gossiping (AAG), e verificando a integridade dos fragmentos salto a salto (hop-by-hop) usando um mecanismo de Compartilhamento de Segredos Verificável baseado em Hash (VSS).

Para avaliar a resiliência desta arquitetura, foram conduzidas simulações distribuídas baseadas em Docker através de topologias lineares, em anel, em malha (mesh) e de gargalo (bottleneck), variando de 10 a 20 nós. Um foco principal da avaliação foi um mecanismo de revogação probabilística, que utiliza um comando "anti-rumor" artificialmente acelerado, projetado para ultrapassar e neutralizar fragmentos em trânsito. Os resultados experimentais demonstraram que essa revogação de via rápida impede com sucesso que os nós alvo reconstruam o segredo, desde que a rede possua profundidade de propagação suficiente. No entanto, a avaliação também expôs uma falha crítica em configurações densas, identificada como a Vulnerabilidade de Convergência de Múltiplos Caminhos (Multipath Convergence Vulnerability). Em estruturas de rede altamente conectadas ou com gargalos, os filtros de roteamento determinísticos afunilam organicamente canais criptográficos distintos para nós intermediários únicos, permitindo que eles acumulem não intencionalmente o limite necessário e reconstruam o segredo prematuramente. Essas descobertas sublinham a necessidade de os sistemas futuros integrarem detecção dinâmica de gargalos, matrizes de reputação descentralizadas e ofuscação de metadados para garantir totalmente os ambientes descentralizados.

Introduction

In a decentralized system, the secure transmission of valuable data becomes an important requirement. When failure or leakage of secret communication happens, it leads to a loss of security, because all intermediate nodes cannot be inherently trustworthy. The entirety of the network's integrity is compromised. As distributed architectures such as blockchain, IoT, or mobile ad-hoc networks (MANET) [4, 9, 12] devices become standard in modern infrastructure, the consequences of exposing cryptographic secrets during peer-to-peer communication are a severe risk.

Epidemic algorithms are widely used to distribute computing due to their robustness, reliability, and fault-tolerance. For example, the Gossip Protocol is highly efficient in its round complexity. Spreading a message within an n -node topology approximately takes $O(\log n)$ rounds of periodic exchanges. However, the major drawback in this protocol is the extreme redundancy of messages, since each node receives the same message multiple times increases network overhead. Hence, this noise consumes bandwidth, consequently creating restrictions on maximum payload capacity [3]. As a result, spreading large cryptographic shares is not only costly but also raises confidentiality concerns, as sensitive data is unnecessarily exposed to intermediate nodes.

There has been a gap since then, where there has been no model to separate the cryptographic payload from the network routing layer. In this context, all intermediate nodes are assumed to be malicious. Current approaches force one to trade off between payload size, network overhead, and strict confidentiality of the data.

To achieve these goals and bridge this gap, we introduce an architecture that completely decouples the cryptographic payload from the gossip routing layer using content-addressable storage, where it can be securely handled without exposing sensitive information. The system utilizes the InterPlanetary File System (IPFS) as a secure and independent storage layer. Shamir's Secret Sharing (SSS), a cryptographic method used to divide a secret into unique fragments, is applied exclusively to the resulting fixed-size Content Identifier (CID), which acts as a unique cryptographic address for data retrieval. Next, these fragments are shared across the network with the use of an adaptive gossip protocol.

Our aim in this study is to explore whether content-addressable storage could be used in a layer that is independent of the payload in a gossip network. Our hypothesis here is that it will lead to significant gains in efficiency. Hypothesis 1 states that isolating SSS to the CID will significantly reduce message redundancy without degrading network coverage time, defined as the total time required for a broadcasted message to successfully propagate to all nodes within the network topology. Finally, Hypothesis 2 states that a propagated revocation command will successfully suppress secret reconstruction, provided it reaches the target before they collect a threshold of valid shares. The tasks to achieve this aim are as follows:

- Investigate the state-of-the-art in scientific writing.
- Simulate a resilient masterless network, using a 2-hop gossip protocol.
- Integrate IPFS as a secure payload layer.
- Apply Shamir's Secret Sharing (SSS) to IPFS CIDs.
- Design and evaluate a revocation mechanism.

The remainder of this work is structured as follows. Section 1 provides the background on gossip protocols and reviews recent advancements in the technology. Section 2 details the design

of the proposed application, followed by its implementation in Section 3. Finally, Section 5 presents the experiments and discusses the evaluation of the system.

1. Related Work

Current approaches to resolving the limitations of gossip protocol can be categorized into different domains: network efficiency, optimization, overhead reduction, and cryptographic secret distribution. In addition, this section reviews the use of decentralized content-addressable storage. By examining these areas, we identify the persistent limitations of routing large cryptographic payloads through untrusted intermediate nodes.

In the domain of network optimization, Adaptive Gossip Protocols [11] allow nodes to alter their communication policies depending on the environment. For example, a protocol could adjust transmission rates according to the weakest node in the network. The main objective is to enhance survivability while minimizing redundant packets. In other research work, gossip is utilized as a probabilistic model. For example, the 2-hop neighbor information where each node calculates a forwarding probability based on the number of neighbors that could also cover those children. This protocol achieves 100% coverage with a higher efficiency rate [1]. In probabilistic models, we had deterministic algorithms based on geolocation and directional antennas [4], which make the routing process overly complex. This research applies the 2-hop neighborhood approach to reduce network overhead.

In terms of communication security, controlling this redundancy prevents broadcast storms and malicious faults. But research has integrated cryptographic techniques directly into gossip. Emphasis is put on increasing confidentiality through fragmentation. For instance, the confidential Gossip algorithm suggests fragmentation of a secret into several pieces utilizing Shamir's Secret Sharing (SSS) [6], allowing data reconstruction even under the loss of some fragments and node crashes.

Moreover, disjoint paths rely on routing confidentiality. Through transmitting data fragments along different and unrelated routes, this approach raises the barriers for an attacker to be able to capture enough packets to compromise the secret [5]. Recent work merges path separation with secret fragmentation, which differs fundamentally from threshold-based SSS by sending each part via a different path [13]. This approach avoids the computational overhead created by cryptographic key management but has some limitations, namely restrictions on payload and bandwidth consumption, and there is no inherent mechanism for secure data revocation.

To overcome the payload and bandwidth limitations when routing large cryptographic payloads requires a shift from traditional data routing to decentralized content-addressable storage [2]. The InterPlanetary File System (IPFS) is a peer-to-peer protocol for storing and sharing decentralized data files. The data is stored locally on computers and identified by cryptographic hash keys known as Content Identifiers (CID). Since addressing only occurs based on the contents alone, IPFS has an inherent property of immutability and duplication elimination of files [7].

Furthermore, in fields ranging from IoT device management [8] to blockchain-based data sharing [10], researchers are leveraging these decentralized models to guarantee highly efficient, secure, and reliable networks when transitioning implemented scenarios into practical, real-world setups.

2. Methodology

Decentralized networks can be represented as undirected graphs, where routing does not depend on any centralized coordination, but through independent nodes to make the decisions. Mathematically speaking, such networks are defined by graphs $G = (V, E)$, where V refers to the set of untrusted intermediate nodes, while E corresponds to the set of communication links between them. The main goal of the design presented is to deliver a secret message from an initial node $n_{org} \in V$ (referred to as Node A), to a destination node, $n_{target} \in V$ (referred to as Node B).

Traditional gossip protocols have restrictions regarding payload capacity. When a network is forced to transmit large encrypted payloads directly in the routing message, every intermediate node has to repeatedly download, decrypt, and re-transmit the data.

To address this limitation, our design solution consists of separating the payload from the routing layer. The idea is to store the data outside of the routing layer, ensuring the graph G is responsible for handling lightweight instructions.

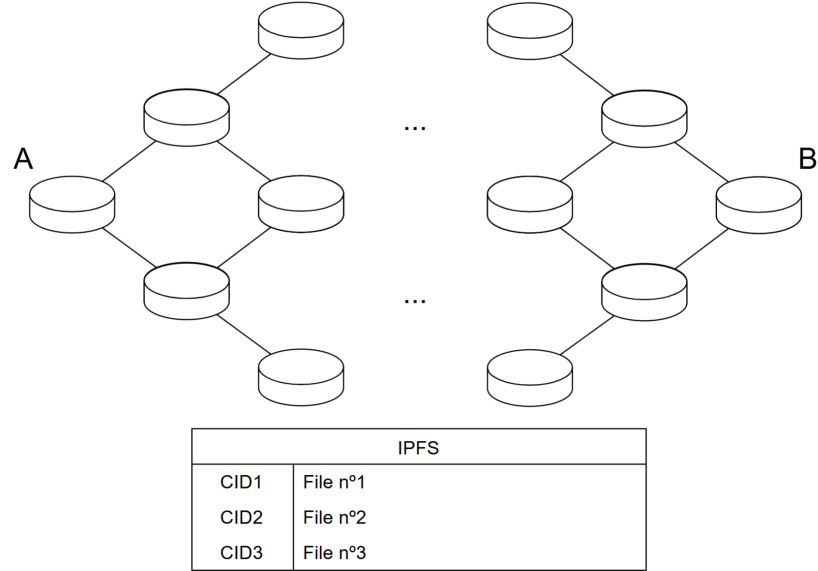


Figure 1: Representation of the decentralized network architecture, illustrating the off-path payload storage and the routing from Node A to Node B.

As illustrated in Figure 1, Node A initiates the message and distributes it through the intermediary gossip mesh, which typically operates as a partial mesh rather than a full graph, until it reaches the target node, Node B. Meanwhile, the actual encrypted payload is stored off-path in an external content-addressable storage layer such as the InterPlanetary File System. Consequently, the data transmitted is limited to the fixed-size IPFS Content Identifier.

The operational process of the system within this architecture relies on three primary mechanisms: Adaptive Secret Sharing, 2-hop neighborhood Gossip, and probabilistic Revocation.

Instead of encrypting and sharing the whole data object, SSS is applied exclusively to the CID. In this regard, the value of k in relation to n varies dynamically depending on the size of the network in order to strike the best balance between redundancy and security. By fragmenting the CID, the bandwidth consumption is reduced without compromising the sensitive data. The secret cannot be retrieved from IPFS until Node B successfully collects the minimum of k required shares.

To avoid duplicate packet transmissions and achieve efficient coverage without flooding the network, the communication process utilizes an Advanced Adaptive Gossiping (AAG) protocol [4]. Through the exchange of neighborhood tables, every node maintains an internal map of its immediate 2-hop neighborhood.

Instead of statically routing data, an intermediate node calculates a dynamic forwarding probability for a received packet based on the availability of redundant routes. The protocol ensures that nodes with fewer connections forward messages with a higher probability to guarantee total network coverage, whereas highly connected nodes forward with lower probability to reduce bandwidth overhead. The precise forwarding logic is detailed in Algorithm 1.

Formally, let c_i be a child node that has not yet received the message, and let $|\text{parent}(c_i)|$ denote the cardinality of the parent set (the number of shared 1-hop neighbors capable of relaying the message to c_i). The forwarding probability P_i for a child c_i is calculated using the established AAG formula derived from Burmester [4]:

$$P_i = 1 - (1 - \tau_{rel})^{\frac{1}{|\text{parent}(c_i)|}}$$

where $\tau_{rel} = \tau_{ARP}^{1/\delta}$, with δ representing the network diameter and τ_{ARP} acting as the target reliability constant (configured to 0.99 in this system). If Node X determines it is the sole available relay ($|\text{parent}(c_i)| = 1$), the equation resolves to $P_i = 1.0$. The final probability bound for routing the packet is determined by $P = \max(P_1, \dots, P_n)$.

Algorithm 1 Advanced Adaptive Gossiping (AAG) Forwarding

Require: Incoming message m from source set S , Node X

```

0:  $\text{child}(X) \leftarrow \{n \in \text{neighbor}(X) \mid n \notin \text{neighbor}(S)\}$ 
0: if  $\text{child}(X) = \emptyset$  then
0:   return {Drop duplicate packet}
0: end if
0:  $P_{max} \leftarrow 0$ 
0: for each  $c_i \in \text{child}(X)$  do
0:    $\text{parent}(c_i) \leftarrow \{p \in \text{neighbor}(c_i) \mid p \in \text{neighbor}(S)\}$ 
0:   Calculate  $P_i$  using AAG probability formula
0:   if  $P_i > P_{max}$  then
0:      $P_{max} \leftarrow P_i$ 
0:   end if
0: end for
0: if  $\text{random}() \leq P_{max}$  then
0:   Forward  $m$  to all  $c_i \in \text{child}(X)$ 
0: end if

```

Applying this reasoning to Figure 1, when a CID share leaves Node A (n_{orig}), the protocol routes the fragments across intermediary nodes toward Node B (n_{target}) using this deterministic disjoint-path probability algorithm.

To ensure the payload remains secure in a zero-trust environment, the system utilizes a prioritized revocation mechanism. If the network detects a breach, Node A broadcasts a REVOKE command (the "anti-rumor"). This command completely bypasses the AAG probability computations ($P = 1.0$) and is forwarded with minimal latency, ensuring a propagation speed advantage

over standard data fragments. When this command arrives at an intermediate node, it immediately purges any locally stored shares and halts further forwarding, starving the network of the k threshold necessary to reconstruct the secret.

3. System Implementation

The system architecture, as illustrated in the component diagram in Figure 2 consists of five interacting components as follows: Docker Network simulation, routing logic in Go, the IPFS storage layer, cryptographic Secret sharing with Verifiable Secret Sharing (VSS), and an automatic revocation process. The following paragraphs provide an elaboration on these components one by one.

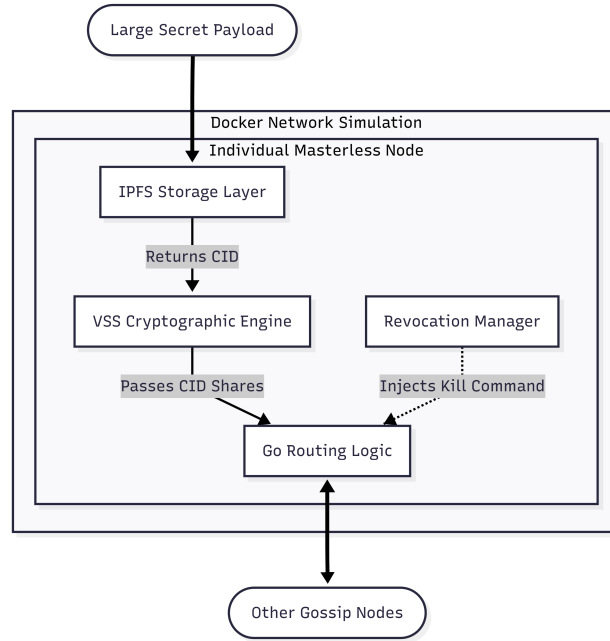


Figure 2: Component diagram of the proposed system architecture, illustrating the internal modules of a single masterless node and its interactions with the IPFS storage layer and the broader gossip network.

To validate the proposed theoretical architecture, a distributed simulation was implemented. The model operates under an active threat model, assuming the intermediary nodes may manipulate the payload messages. However, an adversary cannot manipulate more than half of the nodes of the network. To simulate the network $G = (V, E)$, containerization is applied with Docker. The network consists of 10 to 20 nodes configured in a random graph topology. Every node works as an isolated container communicating through a custom gossip-net bridge network interface. To define the graph structure without adding the discovery step of the network, the topology will be defined during runtime through a JSON configuration file (topology.json). The files utilize a simple adjacency list schema (e.g., "node_id": ["neighbor_1", "neighbor_2"]), based on which every container defines its own 2-hop spatial awareness network.

The routing implementation logic is coded in Go. In order to avoid network deadlocks when dealing with high throughput in the message passing phase, communication between nodes is done using a RESTful HTTP API based on buffered channels. When a node gets a gossip message, the data gets written into the buffered channel, where a separate go-routine reads and handles the

cryptography verification, state updates, and caching. Moreover, once the node computes the AAG probabilities and chooses to transmit a fragment to its children, it makes an HTTP request asynchronously.

To enforce the separation between the cryptographic payload and the routing layer, a localized InterPlanetary File System (IPFS) node (Kubo) is deployed alongside the gossip protocol. When a node n_{orig} tries to distribute its secret, it uses the local IPFS HTTP API to submit the payload and receives a unique, fixed-size Content Identifier (CID). To preserve bandwidth capacity, only the CID is propagated through the network. Figure 4 illustrates the system component architecture, highlighting the data flow from n_{orig} through the Go-based routing layer, its interaction with the IPFS daemon, and concluding with the VSS verification step at n_{target} .

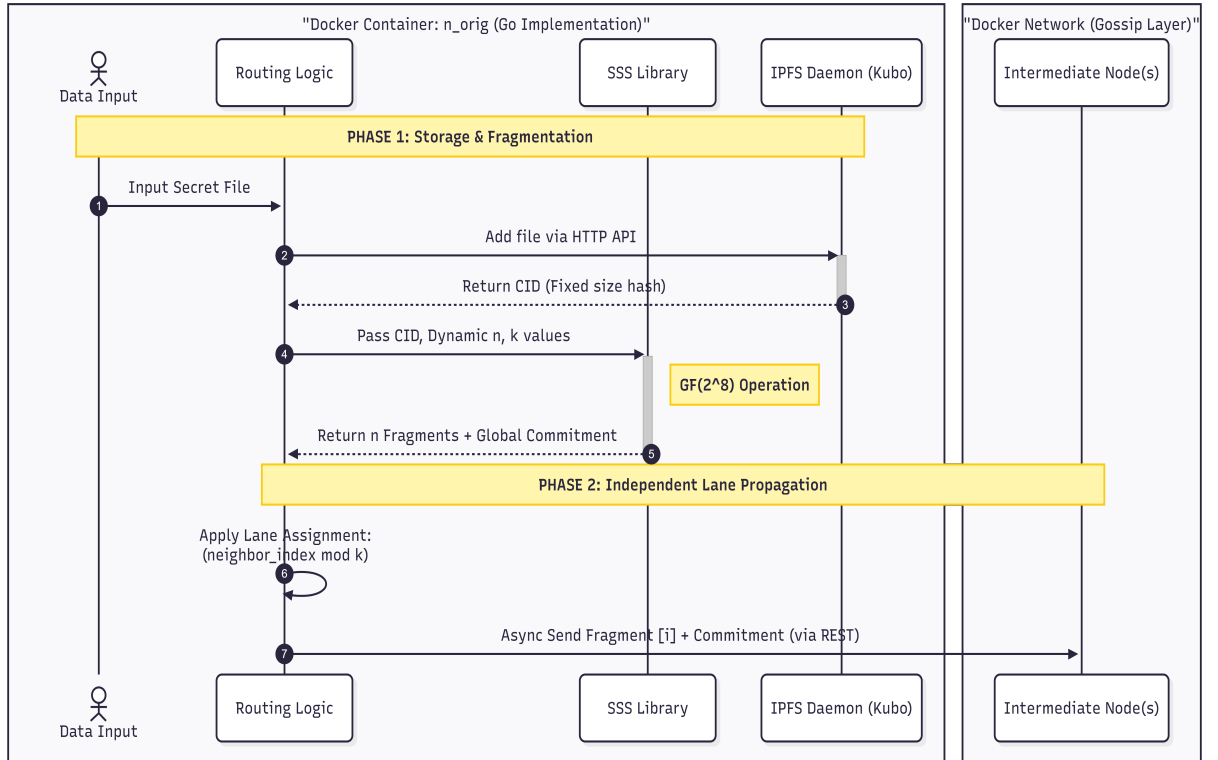


Figure 3: Sequence diagram illustrating the system architecture and payload data flow, specifically detailing the initial IPFS ingestion and Shamir’s Secret Sharing (SSS) fragmentation at the originator node (n_{orig}).

The k -out-of- n Shamir’s Secret Sharing (SSS) algorithm is integrated using the github.com/corvus-ch/shamir library (version [1.0.1]), operating over the Galois Field $GF(2^8)$. Instead of static fragmentation, the system dynamically calculates the values for both the number of shares (n) and the necessary reconstruction threshold (k), based on the localized network density of the originator node. Because of that, the total amount of shares (n) will be the same as the quantity of neighbors of the originator node, while the threshold will be determined as a strict network majority ($n/2 + 1$).

To enforce the disjoint pathing due to the nature of the routing model, the implementation utilizes dynamic deterministic filters in the AAG forwarding logic. Fragment transmission is done by applying a lane assignment function defined as $L_i = \text{neighbor}_{index} \pmod k$. This is a mathematical restriction that ensures that the fragments are forced into distinct paths. For instances where there are duplicates, i.e., when n cannot be evenly divided by k , the additional shares will go down other routes determined by a random lane assignment. The sharing process requires protection from any

possible tampering on these untrusted routes; the implementation integrates a Hash-Based VSS for security purposes. During the fragment forwarding process, intermediate hops calculate the hash of the message and compare it with the hash value included in the share. If the two hashes do not match, the packet is dropped. Once the node n_{target} gathers the requisite k shares, verification of the CID takes place with respect to the commitment broadcast by n_{orig} .

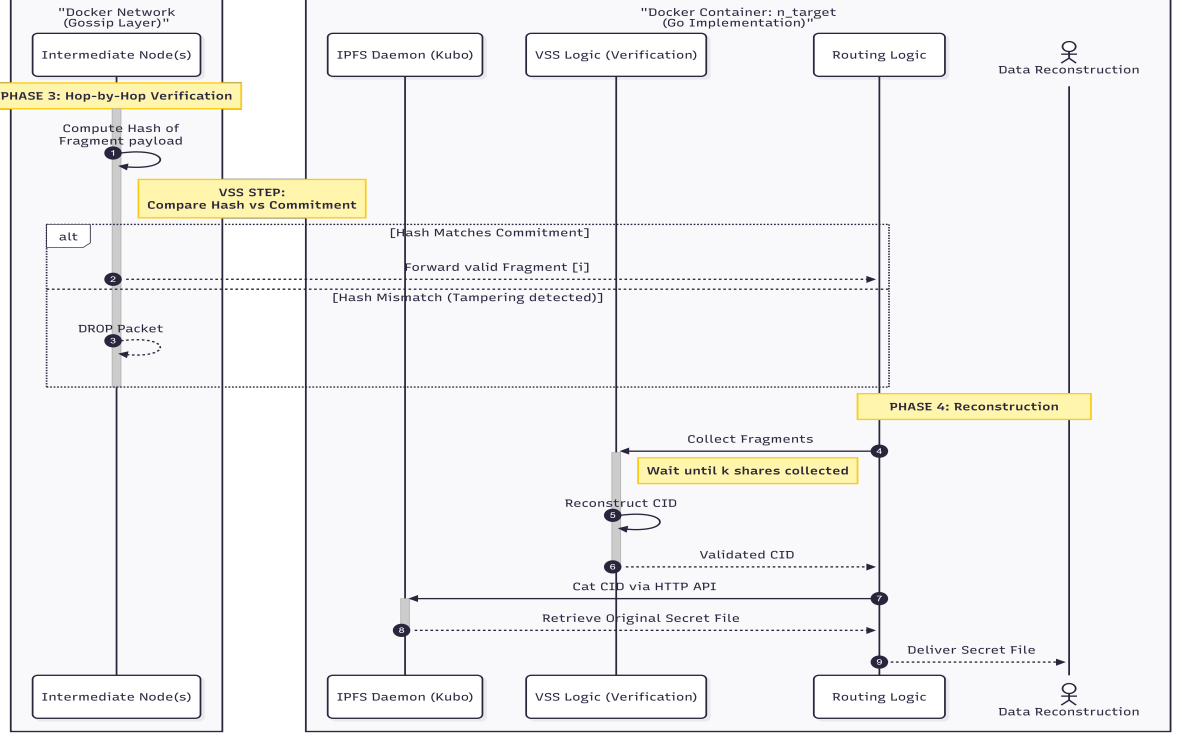


Figure 4: Sequence diagram illustrating the system architecture the hop-by-hop Verifiable Secret Sharing (VSS) validation across intermediate Docker-based network nodes, and the final reconstruction and retrieval at the target node (n_{target}).

The revocation mechanism uses propagation speed differentials, utilizing simulated delays for the network packets. Standard CID fragments incorporate an artificial processing delay of 200 milliseconds per hop. This specific baseline latency was introduced to establish a distinct temporal window, thereby facilitating granular log analysis, deterministic event-ordering verification, and clear behavioral observability across the distributed containers during the simulation phase. When a node receives a revoke command, it is recognized by the unique fragment ID. The protocol bypasses all AAG probability computations and transmits the anti-rumor to all direct neighbors in 10 milliseconds to prevent race conditions, without the delay CPU processes the broadcast too fast. This creates a priority propagation channel, enabling the revocation command to autonomously overtake transit fragments before downstream nodes can successfully accumulate the required k shares.

4. Experiments

To evaluate the dissemination behavior and revocation mechanism proposed in this study, three different routing topologies were created with a node range of 10 to 20 nodes. A linear topology with 10 nodes was chosen to ensure a uniform progression of the dissemination and a singular

forwarding route. A ring topology with 10 and 20 nodes was included to introduce multi-path propagation, enabling the anti-rumor to traverse the network in different directions. A highly connected mesh topology with 15 and 20 nodes was included to simulate realistic, dense ad-hoc scenarios. Lastly, a bottleneck topology with 10 and 15 nodes was specifically designed to evaluate the system against the Multipath Convergence Vulnerability by artificially forcing disjoint paths to intersect at a single intermediate routing node. The evaluation was done by assessing the dissemination and reconstruction behavior under various Secret Sharing setups, where the number of shares was either 2, 3, or 4, and the number of thresholds was either 2 or 3.

System behavior was measured using a structured analysis of event logs produced by nodes in a decentralized manner. We identified and tracked four types of event logs: receiving a CID fragment, receiving an anti-rumor, forwarding operations, and reconstruction. The evaluation criteria of the system can be defined using a binary measure according to the events that occur. A successful revocation is defined as the explicit absence of a "reconstruction attempt" event across all node logs, indicating that the anti-rumor successfully starved intermediate and target nodes of the threshold (k). Conversely, any occurrence of a "reconstruction attempt" event prior to the arrival of the anti-rumor constitutes revocation failure. Because the simulation utilizes concurrent Docker execution, cryptographically randomized Shamir polynomials, and probabilistic AAG forwarding, the precise propagation paths are inherently non-deterministic. To account for this, the results presented for each topology represent the dominant behavioral trend observed across multiple independent simulation runs.

In order to conduct a detailed quantitative analysis of the temporal dynamics of the system, Figure 5 presents the time analysis of the revocation race condition on different network topologies. The graph below compares the values of two key temporal parameters that can be obtained from the decentralized logs: Secret Convergence ETA (estimated time for obtaining k valid shares of the payload) and Detection Delay (probabilistic latency until the breach is detected by the originator node and the prioritized anti-rumor is injected). As can be seen from the graph, highly connected networks, such as the Mesh topology, Bottleneck topology, and the extended Ring topology, show extremely fast convergence time and it always falls between 600 ms and 800 ms.

Notably, in all tested multipath configurations, this time of convergence drops substantially lower than the threshold detection delay (consistently above 900ms for denser networks). As a result, by the time that the anti-rumor message is injected into the network, the cryptographic threshold (k) has long since been surpassed in the downstream nodes. Such empirical evidence supports the Multipath Convergence Vulnerability: although the revocation procedure enjoys an accelerated propagation speed (modeled at 10ms hops), this speed-up advantage is rendered moot when the secret converges more quickly than the initial breach detection time through network density.

5. Results

Due to the non-deterministic nature of the system, individual simulation runs may produce different propagation paths and fragment identifiers. The findings reported below therefore reflect the consistent behavioral pattern observed across repeated executions of each topology configuration.

The first result of the experiment highlights the important relationship between fragment propagation speed and revocation timing. The purpose of this evaluation was to prove that an anti-rumor can travel faster than standard fragments, purging them in advance to secure the system by rendering reconstruction impossible. As observed in Table ??, the success of the revocation heavily

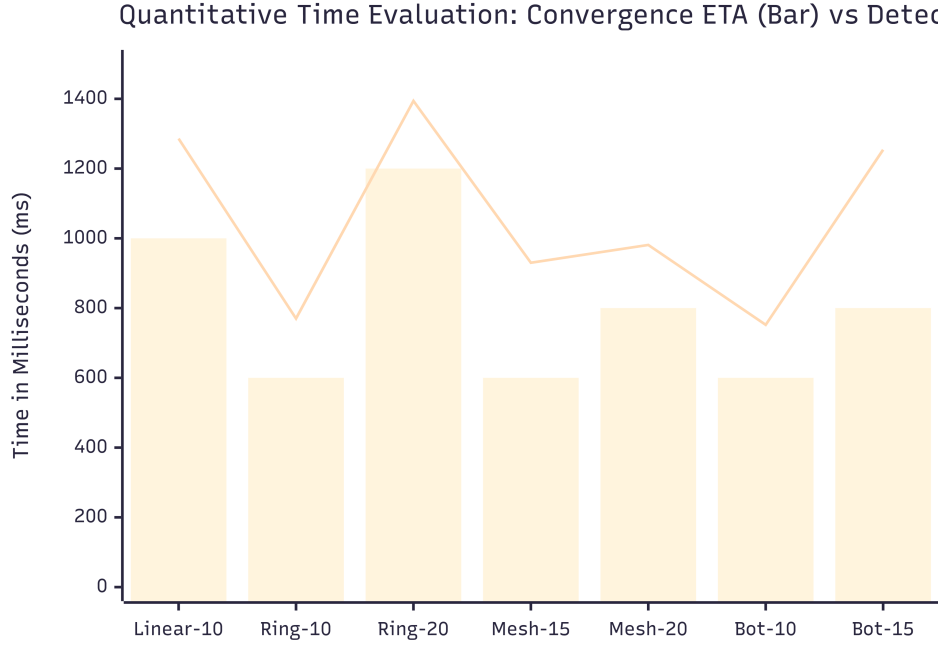


Figure 5: Quantitative Time Evaluation of the Revocation Race Condition based on Docker simulation logs. Across all highly connected and multipath topologies, the secret convergence ETA (yellow) falls below the revocation detection delay (red), resulting in the secret successfully outpacing the anti-rumor.

depends on the network depth and the specific topology through which the anti-rumor and the CID fragments are disseminated.

Probabilistic revocation was successful only if the anti-rumor reached the intermediate and target nodes before they had time to collect the required threshold of shares (k). For instance, in the 10-node Ring topology ($n = 2, k = 2$), the revocation attempt was unsuccessful (resulting in 1 node still reconstructing the secret), because the network was too shallow for the anti-rumor to overtake the fragments. In contrast, expanding the Ring topology to 20 nodes provided sufficient network depth, allowing the revocation to successfully block all reconstructions (0 nodes). Furthermore, highly connected configurations—such as the Mesh and Bottleneck topologies—demonstrated that multipath convergence allows intermediate nodes to rapidly gather the k threshold before the anti-rumor can arrive, causing the revocation to fail and resulting in multiple intermediate nodes reconstructing the secret.

However, these findings reflect an underlying contradiction in relation to network robustness versus revocation capability. Mesh and Bottleneck have long been considered popular topology types for masterless systems due to their fault tolerance and rapid spread of information. Yet, our results show the contrary – dense topologies prove to be the worst fit for temporal revocation solutions. This is due to the way Shamir’s Secret Sharing works with dense topologies since reconstruction needs only a certain number of shares (k). Thanks to multipath convergence, intermediate nodes within the dense network get those shares even before a prioritized anti-rumor has had the chance to reach those nodes hop by hop and cancel the secret dissemination.

From the practical perspective, such findings should be taken into account in the design of future zero-trust decentralized storage systems. This shows that speed of propagation in dense topologies alone cannot serve as an adequate revocation method. Instead, the threshold (k) of SSS should dynamically depend on local network density to give the anti-rumor enough time to cancel the

secret.

All simulations were conducted on a single host machine running Windows 11, utilizing the Windows Subsystem for Linux (WSL2) with an Ubuntu distribution. The physical hardware environment was equipped with an Intel(R) Core(TM) Ultra 9 285H and 32.GB of RAM. Within this virtualized environment, each network node was instantiated as an isolated Docker container running Go v1.21 and interconnected via a custom bridge network. The IPFS storage layer was managed by a local Kubo v0.24.0 instance. No external network traffic was involved in any simulation run.

Topology	Nodes	n	k	Revocation	Int. Recon.	Secret Reconstructed
Linear	10	2	2	No	No	1
Linear	10	2	2	Yes	No	0
Ring	10	2	2	No	No	1
Ring	10	2	2	Yes	No	1
Ring	20	2	2	No	No	1
Ring	20	2	2	Yes	No	1
Mesh	15	3	2	No	No	7
Mesh	15	3	2	Yes	No	1
Mesh	20	4	3	No	No	2
Mesh	20	4	3	Yes	No	1
Bottleneck	10	3	2	No	Yes	4
Bottleneck	10	3	2	Yes	Yes	2
Bottleneck	15	4	3	No	Yes	2
Bottleneck	15	4	3	Yes	Yes	2

Table 1: Experimental setup and binary evaluation of the Probabilistic Revocation Mechanism across varying topologies, highlighting Multipath Convergence vulnerabilities.

6. Discussion

In this study, the main contribution of the research is the analysis of the proposed architecture that decouples cryptographic payloads from the routing layer using a decentralized content-addressable storage (IPFS), dynamic Shamir’s Secret Sharing (SSS), and Advanced Adaptive Gossiping (AAG). The primary finding is that the probabilistic revocation approach can indeed intercept and render in-transit secrets effectively. As observed in the Mesh topology with 15 nodes, propagation speed variation was introduced, allowing the anti-rumor to overtake the fragment payload, preventing the accumulation of the required threshold k and rendering reconstruction impossible.

On the other hand, the experimental evaluation also exposed some limitations. In dense centralized network structures, particularly in the bottleneck topology, a flaw was exposed because of the multi-path convergence. Even though the algorithm adopts a deterministic modulo filter approach to ensure disjoint routing paths, physical network bottlenecks naturally funnel distinct cryptographic channels. Consequently, these intermediate nodes unintentionally end up with the k threshold and are able to reconstruct the secret.

This structural weakness creates an attack surface for adversaries with knowledge of the network topology. A malicious node positioned within a bottleneck will be able to intercept the requisite number of fragments needed to reach the threshold, without any possibility of detection. The hash-based VSS being utilized in our system assures the integrity of fragments, but does not prevent their convergence at bottleneck nodes. These identified limitations form the basis for future research on topology awareness and malicious node isolation. To overcome the constraints of bottleneck topologies, the next version of the AAG algorithm must take into account dynamic bottleneck detection. Through the capability of nodes to monitor network density dynamically, AAG can then use variable forwarding probabilities that will help it avoid routing shares through the points of high centralization. As far as the problem of replay and replication attacks is concerned, future research should pay attention to decentralized reputation matrices that will help it detect, punish, and circumvent nodes who tend to replicate and/or retain fragments excessively. Moreover, using some metadata obfuscation technologies, such as Onion Routing, in the gossip layer will help to eliminate multipath convergences completely.

Conclusions and Future Work

This study demonstrated an architecture that addresses the competing demands of redundancy, bandwidth efficiency, and confidentiality in decentralized networks. By decoupling the cryptographic payload from the routing layer through IPFS, applying dynamic Shamir’s Secret Sharing exclusively to the fixed-size Content Identifier, and disseminating the resulting shares via the AAG protocol, the system successfully minimizes network overhead without exposing sensitive data to intermediate nodes. The experimental results further confirmed that a probabilistic revocation mechanism, implemented as a priority anti-rumor command with a structural propagation speed advantage, can prevent reconstruction in topologies with sufficient depth. However, the identification of the Multipath Convergence Vulnerability in dense and bottleneck topologies establishes a clear boundary for the current system, as structural convergence points allow intermediate nodes to unintentionally accumulate the required threshold, causing revocation to fail.

Future research must address these limitations through three directions. First, the AAG protocol should incorporate dynamic bottleneck detection, allowing nodes to monitor local network density and adjust forwarding probabilities in real time, transforming the current static modulo assignment into a topology-sensitive mechanism. Second, the integration of decentralized reputation matrices would allow the network to identify and route around nodes exhibiting suspicious behavior, such as excessive fragment retention or duplication. Third, the application of metadata obfuscation techniques at the gossip layer, such as Onion Routing, could prevent intermediate nodes from correlating fragments belonging to the same CID, neutralizing the informational advantage that bottleneck positioning currently provides to a potential adversary.

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